Birla Vishvakarma Mahavidyalaya

Information Technology Department
(An Autonomous CVM Institute)
Vallabh Vidyanagar

Subject: 4IT31 (Project-I), AY: 2026-27

Project Evaluation Report

Group	09
Project Title	Chess Playing Robotic Arm
Faculty Guide	Dr. Keyur Brahmbhatt
Report	Progress Report - 2
Date of Evaluation	

Group Members

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Comment:

Faculty Guide Signature

1. System Diagram

1.1 Activity Diagram

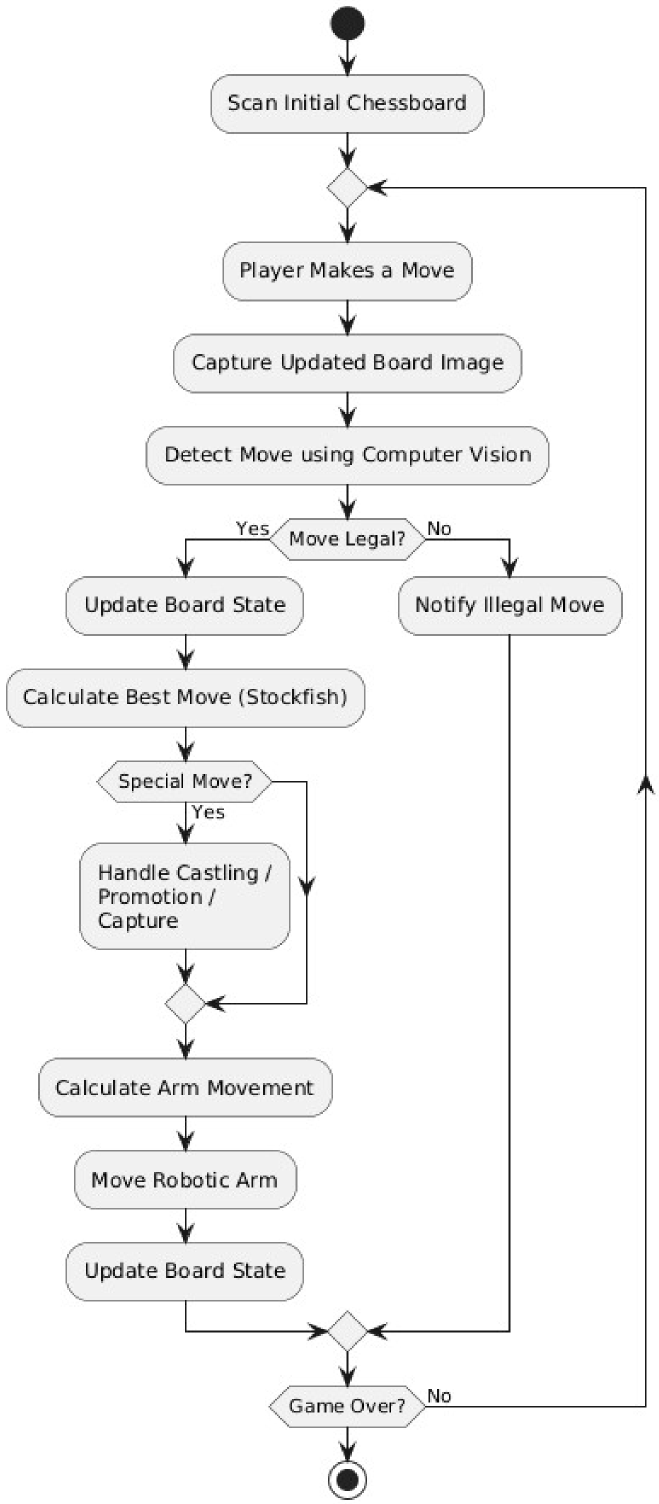

Figure 1.1 Activity Diagram

1.2 Sequence Diagram

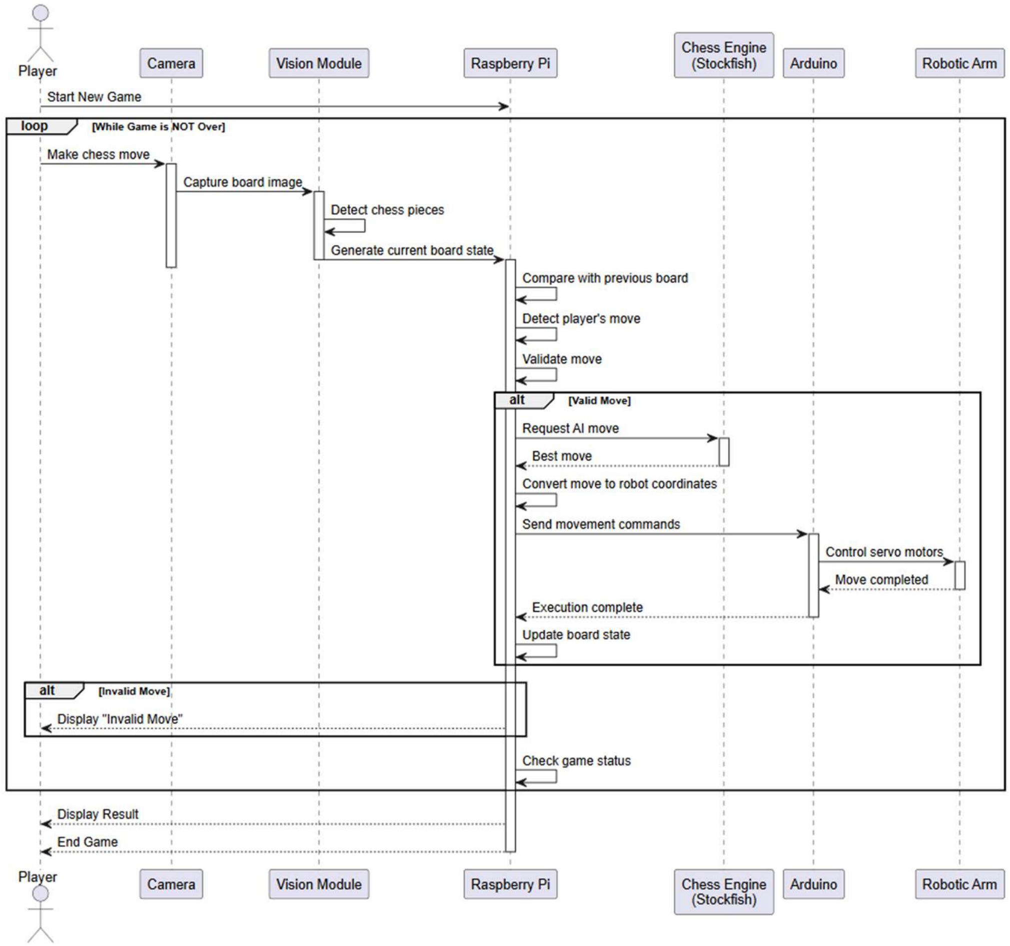

Figure 1.2 Sequence Diagram

1.3 Hardware Circuit Diagram

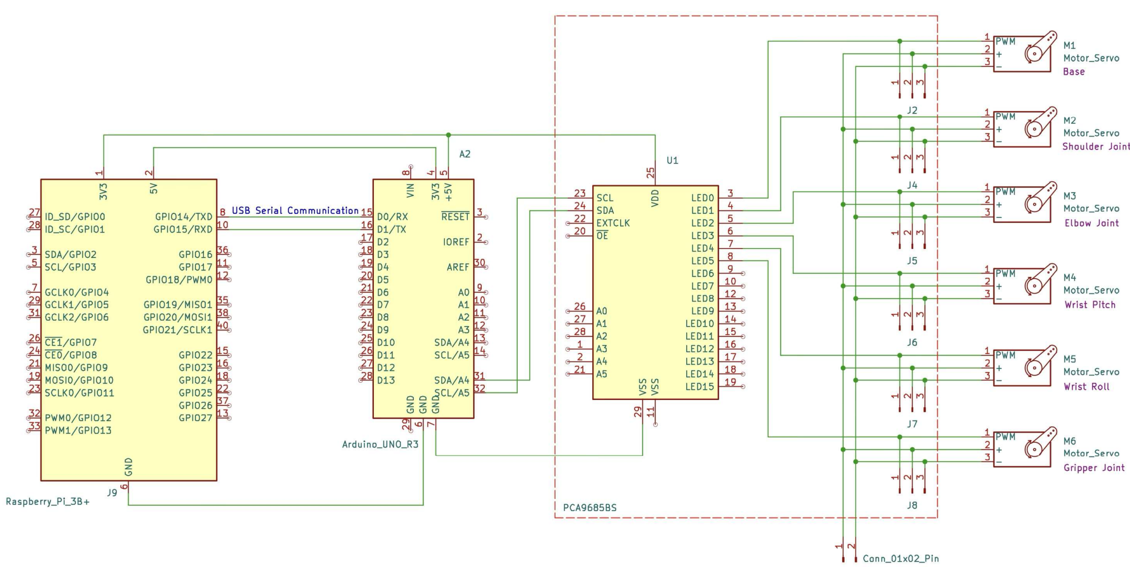

Figure 1.3 Hardware Circuit Diagram

1.4 Data Flow Diagram

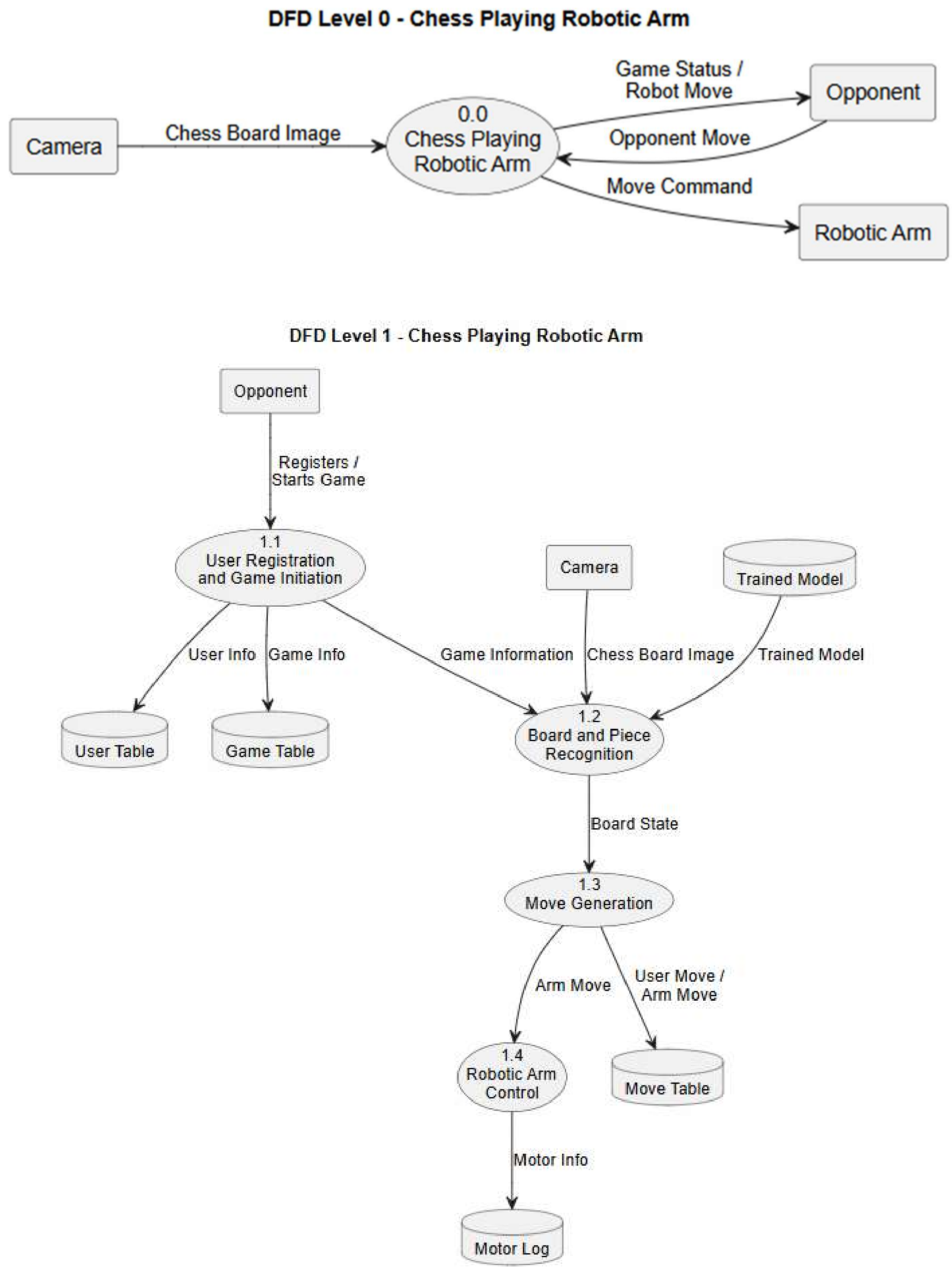

Figure 1.5 Data Flow Diagram

2. Module Specifications

2.1 Camera-Based Board Monitoring

This module captures images of the chessboard using a top-mounted camera and processes them to detect the board and the positions of all chess pieces. The processed board state is then provided to the move detection module.

2.2 Move Detection Module

This module compares consecutive board states to identify the player's move and validates it according to the rules of chess before forwarding it to the chess AI engine.

2.3 Chess AI Engine

This module uses the Stockfish chess engine to analyze the current board position and generate the optimal move based on the selected difficulty level while ensuring all chess rules are followed.

2.4 Coordinate Mapping Module

This module, running on the Raspberry Pi, converts the chess engine's move into physical coordinates, computes the required joint angles using inverse kinematics, and sends the corresponding servo angle commands to the Arduino Uno.

2.5 Robotic Arm Control Module

This module receives servo angle commands from the Raspberry Pi and controls the robotic arm by generating PWM signals for the servo motors, enabling accurate pick-and-place operations on the chessboard.

2.6 Game Management Module

This module manages the overall game flow by maintaining the game state, coordinating communication between all modules, handling player turns, and ensuring smooth execution of the chess game.

3. Technology Stack Selection

3.1 Software

- Programming Language: Python, C
- Game Interface: HTML, CSS, JavaScript
- Operating System: Raspberry Pi OS
- Computer Vision Library: OpenCV
- Chess Engine: Stockfish

- Communication Protocol: Serial UART
- Arduino IDE for motor control programming
- Database: PostgreSQL

3.2 Hardware

- Raspberry Pi 3B+
- Arduino UNO R3
- 6-DOF Robotic Arm
- Servo Motors – SG 1501MG
- Servo Driver – PCA9685
- Camera/Camera Module
- Camera Stand
- Miscellaneous: External Power Supply, Jumper Wires, etc.

4. Literature Review

4.1 Research Papers

4.1.1 Automated Chess Gameplay With Computer Vision and a Robotic Arm (AlKhatib et al., 2026, *Engineering Reports*)

Link: https://onlinelibrary.wiley.com/doi/full/10.1002/eng2.70607

Integrates YOLOv8, Stockfish, and a robotic arm into a unified autonomous chess-playing system. Uses FEN representation and robotic motion planning to execute moves.

4.1.2 Autonomous Chess Playing System with Computer Vision and Collaborative Robots (ICARA 2026)

Link:

https://www.researchgate.net/publication/404057428_Autonomous_Chess_Playing_System_with_Computer_Vision_and_Collaborative_Robots

Develops a chess-playing robot using computer vision, a collaborative robotic arm, ArUco marker calibration, and the Stockfish engine for autonomous gameplay.

4.1.3 Development of an Autonomous Chess Robot System Using Computer Vision and Deep Learning (Truong & Bui, 2025, *Results in Engineering*)

Link:

https://www.researchgate.net/publication/388307913_Development_of_an_Autonomous_Chess_Robot_System_Using_Computer_Vision_and_Deep_Learning

Presents a complete low-cost chess robot using a 4-DOF SCARA robotic arm, deep learning for chess piece recognition, computer vision for board state detection, and the Stockfish engine for move generation.

4.1.4 An Open-Source Reproducible Chess Robot for Human–Robot Interaction Research (Zhang et al., 2024)

Link: *https://www.frontiersin.org/journals/robotics-and-ai/articles/10.3389/frobt.2025.1436674/full*

Introduces an open-source chess robot capable of recognizing chess pieces, physically moving them, and interacting with users through speech and gestures. The work emphasizes modular software and reproducibility.

4.1.5 Robust Computer Vision Chess Analysis and Interaction with a Humanoid Robot (MDPI Computers, 2019)

Link: *https://www.mdpi.com/2073-431X/8/1/14*

Focuses on robust chessboard and piece recognition using computer vision, while integrating Stockfish and inverse kinematics for robot interaction.

4.2 Existing Systems

4.2.1 EDGE-tronics – Chess-Robot

Link: *https://github.com/EDGE-tronics/Chess-Robot*

A Raspberry Pi-based autonomous chess-playing robot using a 4-DOF robotic arm, OpenCV for board detection, and Stockfish as the chess engine. The system focuses on integrating computer vision with robotic manipulation for physical gameplay.

4.2.2 harirakul – ChessRobotArm

Link: *https://github.com/harirakul/ChessRobotArm*

Implements a robotic arm capable of executing chess moves. The repository mainly focuses on robotic arm motion, inverse kinematics, and move execution.

4.2.3 harshpatel097 – Chess-Playing-AI-Bot

Link: *https://github.com/harshpatel097/Chess-Playing-AI-Bot*

Primarily focuses on chess AI and move generation using computer vision and software algorithms, with limited emphasis on robotic manipulation.

4.2.4 METU-KALFA – chess_mate

Link: *https://github.com/METU-KALFA/chess_mate*

A ROS-based chess robot integrating robot motion planning, perception, and manipulation. It demonstrates coordinated operation between robotic components using ROS.

4.2.5 LuckCow – ChessRobot

Link: *https://github.com/LuckCow/ChessRobot*

An autonomous chess-playing robot combining robotic arm control, computer vision, and chess engine integration to physically play chess against a human.

Existing System	Limitation of the System	How Our System is Different and Better
EDGE-tronics – Chess-Robot	Uses a 4-DOF robotic arm, limiting flexibility and movement precision. The system focuses mainly on integrating computer vision with robotic manipulation	Uses a 6-DOF robotic arm, providing greater flexibility and more accurate pick-and-place operations. Includes dedicated coordinate mapping, inverse kinematics, and modular game management.
harirakul – ChessRobotArm	Primarily focuses on arm motion, inverse kinematics without much focus on computer vision.	Integrates computer vision, move detection, chess AI, coordinate mapping, and robotic arm control into a complete chess-playing system.
harshpatel097 – Chess-Playing-AI-Bot	Concentrates on chess AI and move generation with limited support for physical robotic manipulation.	Combines AI-based move generation with a physical robotic arm, enabling autonomous execution of moves on a real chessboard.
METU-KALFA – chess_mate	Relies on the ROS framework, making the system more complex to configure for beginners.	Uses a simpler Raspberry Pi–Arduino architecture without requiring ROS, making it easier to implement.
LuckCow – ChessRobot	Provides automation but has a less modular architecture	Follows a modular design, making it easier to develop, maintain, and extend.

Table 4.1 Comparison Study with Existing Systems

5. Hardware Requirements

Component	Quantity	Purpose
Raspberry Pi 3B+	1	Runs YOLO model, Stockfish engine, and overall system control
Arduino Uno R3	1	Controls servo motors of the robotic arm
6-DOF Robotic Arm	1	Physically moves chess pieces
Servo Driver PCA9685	1	Controls multiple servo motors efficiently
Servo Motors SR 1501MG	6	Actuate each robotic arm joint
Camera	1	Captures the chessboard image
Camera Stand	1	Mount camera on top of the board
Power Supply	1	Supplies power to Raspberry Pi and robotic arm
Jumper Wires	As required	Electrical connections
Chessboard and Chess Pieces	1 Set	Playing environment

Table 5.1 Parts required

6. Hardware Procurement

6.1 Selection Criteria

Raspberry Pi 3B+

- Quad-core processor suitable for running Python applications.
- Supports OpenCV and YOLO inference.
- Can communicate with Arduino via USB Serial.
- Low power consumption.

Arduino Uno R3

- Generates accurate PWM signals.
- Easy integration with servo motors.
- Widely supported libraries.

6-DOF Robotic Arm

- Sufficient reach to access every square on the chessboard.
- Six degrees of freedom allow flexible positioning.
- Ensuring servo motors provide enough torque.

6.2 Procurement Sources

- Raspberry Pi 3B+

https://quartzcomponents.com/products/raspberry-pi-3-model-b

- Arduino Uno R3

https://robu.in/product/arduino-uno-r3

- 6-DoF Robotic Arm with SR 1501MG Servo Motors

https://robu.in/product/6dof-robotic-arm-multi-degree-of-freedom-robot-servo-grab-maker-education-kit-unassembled

- Servo Driver PCA9685

https://robu.in/product/servo-driver-module-shield-16-channel/

7. Hardware BoM

Component	Comments	Quantity	Price	Total Amount
Raspberry Pi 3B+	-	1	4637	4637
Arduino Uno R3	-	1	420	420
6-DOF Robotic Arm	-	1	8899	8899
Servo Driver PCA9685	-	1	547	547

Servo Motors SR 1501MG	incl. with 6-DoF Robotic Arm	6	-	-
Camera	Using phone camera	1	-	-
Camera Stand		1	542	542
Misc.	Wires, Connectors, Power Supply	-	500	500
Chessboard and Chess Pieces	Available	1 Set	-	-
Total				15545

Table 7.1 Hardware BoM

8. Hardware Specifications

8.1 Raspberry Pi 3B+

- Processor: Broadcom BCM2837B0 Quad-Core Cortex-A53 @ 1.4 GHz
- RAM: 1 GB LPDDR2
- Connectivity: Wi-Fi, Bluetooth 4.2, Ethernet
- GPIO Pins: 40
- Operating System: Raspberry Pi OS
- Role: Image processing, AI execution, Motion planning

Figure 8.1 Raspberry Pi 3B+

8.2 Arduino Uno R3

- Microcontroller: ATmega328P
- Operating Voltage: 5 V
- Clock Speed: 16 MHz
- Digital I/O Pins: 14
- PWM Pins: 6
- Communication: USB, UART, I²C, SPI

- Role: Receives servo angle commands from Raspberry Pi and controls servo motors via the PCA9685 servo driver

Figure 8.2 Arduino Uno R3

8.3 6-DOF Robotic Arm

- Degrees of Freedom: 6
- Material: Aluminium Alloy
- End Effector: Servo-operated gripper
- Horizontal Reach: 300 mm (approx.)

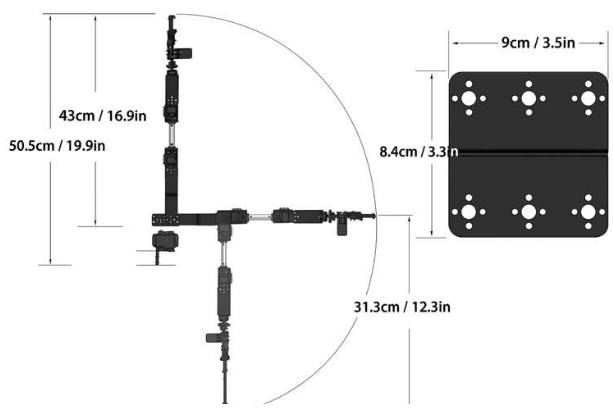

Figure 8.3 Arm reach

- Payload Capacity: 200–300 g (approx.)
- Role: Performs pick-and-place operations of chess pieces on the chessboard

Figure 8.4 Assembled 6 DoF Robotic Arm

Figure 8.5 Unassembled Parts

8.4 Servo Driver PCA9685

- PWM Channels: 16
- PWM Resolution: 12-bit (4096 levels)
- Communication Protocol: I2C
- Logic Voltage: 3.3–5 V
- Servo Supply Voltage: Up to 6 V (external)
- Frequency Range: Up to 1.6 kHz
- Role: Generates PWM signals to control multiple servo motors simultaneously

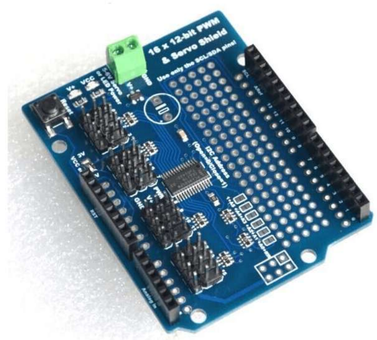

Figure 8.4 Servo Driver PCA9685

8.5 Servo Motors SR 1501MG

- Model: SR1501MG
- Rotation Angle: 180°
- Operating Voltage: 4.8–6.0 V
- Gear Type: Metal Gear
- Control Signal: PWM
- Stall Torque: Approximately 15 kg·cm @ 6 V
- Role: Provides rotational movement to the robotic arm joints for precise positioning and manipulation of chess pieces

Figure 8.5 Servo Motor SR 1501MG

9. Hardware Installation Steps

Referred:

- *https://robu-prod-media.s3.ap-south-1.amazonaws.com/products/attachments/Eq0SH3f4IQXoK1XIIEUwK8F78xItwCXGfefhfO3I.pdf*
- *https://youtu.be/bKY8MJDq3p0?si=Nmn4Yjr43cZ-oSZL*